

Day 5 – MongoDB Assignment

To open MongoDB Shell

```
C:\Users\HP>mongosh
```

Current Mongosh Log ID: 69574e97fae8ace425523741

Connecting to:

```
mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.1.1
```

Using MongoDB: 8.2.3

Using Mongosh: 2.1.1

mongosh 2.5.10 is available for download: <https://www.mongodb.com/try/download/shell>

For mongosh info see: <https://docs.mongodb.com/mongodb-shell/>

The server generated these startup warnings when booting

2025-12-23T09:38:04.003+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted

Create database

```
test> use insuranceDB;
```

switched to db insuranceDB

Create Collections

```
insuranceDB> db.createCollection("customers")
```

```
{ ok: 1 }
```

```
insuranceDB> db.createCollection("policies")
```

```
{ ok: 1 }
```

```
insuranceDB> db.createCollection("policyAssignments")
```

```
{ ok: 1 }
```

```
insuranceDB> db.createCollection("agents")
```

```
{ ok: 1 }
```

```
insuranceDB> db.createCollection("claims")
```

```
{ ok: 1 }
```

Collections in insuranceDB

```
insuranceDB> show collections
```

```
insuranceDB> show collections
agents
claims
customers
policies
policyAssignments
```

Inserting into Customers Collection

```
insuranceDB>db.customers.insertMany([{customerid:1,firstName:"Amit",lastName:"Sharma",dateOfBirth:new Date("1995-10-01"),phone:"99897275",email:"amit@mail.com",address:null,city:null},{customerid:2,firstName:"preethi",lastName:"varma",dateOfBirth:new Date("1998-09-01"),phone:"89898989",email:"preethi@mail.com",address:null,city:null},{customerid:3,firstName:"Vishwa",lastName:"Nair",dateOfBirth:new Date("2014-10-01"),phone:"89765987",email:"vishwa@mail.com",address:null,city:null},{customerid:4,firstName:"Hansy",lastName:"Kamtala",dateOfBirth:new Date("2008-10-01"),phone:"83281187",email:"hansika@mail.com",address:null,city:null},{customerid:5,firstName:"Julien",lastName:"Gou",dateOfBirth:new Date("2000-09-01"),phone:"99897678",email:"julien@mail.com",address:null,city:null}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('695755fcfae8ace42552375a'),
    '1': ObjectId('695755fcfae8ace42552375b'),
    '2': ObjectId('695755fcfae8ace42552375c'),
    '3': ObjectId('695755fcfae8ace42552375d'),
    '4': ObjectId('695755fcfae8ace42552375e')
  }
}
```

Inserting into Agents Collection

```
insuranceDB>
```

```
db.agents.insertMany([{agentid:101,agentName:"Santhosh",phone:"99898989",city:"Hyderabad"},{agentid:102,agentName:"Ravi",phone:"97658987",city:"Delhi"},{agentid:103,agentName:"Girish",phone:"89787897",city:"Hyderabad"},{agentid:104,agentName:"Hithesh",phone:"83383345",city:"Kakinada"}])
```

```
}
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69575647fae8ace42552375f'),
    '1': ObjectId('69575647fae8ace425523760'),
    '2': ObjectId('69575647fae8ace425523761'),
    '3': ObjectId('69575647fae8ace425523762')
  }
}
```

Inserting into Policies Collection

```
insuranceDB> db.policies.insertMany([
  {policyid:120,policyName:"Health
Secure",policyType:"Health",premiumAmount:17569,durationYears:1},
  {policyid:121,policyName:"Life
Shield",policyType:"Life",premiumAmount:25000,durationYears:10},
  {policyid:122,policyName:"Motor Protect",policyType:"Motor",premiumAmount:8000,durationYears:1}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69575663fae8ace425523763'),
    '1': ObjectId('69575663fae8ace425523764'),
    '2': ObjectId('69575663fae8ace425523765')
  }
}
```

Inserting into policyAssignments Collection

```
insuranceDB> db.policyAssignments.insertMany([
  {assignmentid:201,customerid:2,policyid:120,agentid:101,startDate:new Date("2025-10-01"),endDate:new Date("2025-10-01")},
  {assignmentid:202,customerid:1,policyid:120,agentid:101,startDate:new Date("2020-10-01"),endDate:new Date("2026-12-01")},
  {assignmentid:205,customerid:5,policyid:122,agentid:103,startDate:new Date("2025-12-01"),endDate:new Date("2026-12-01")},
  {assignmentid:206,customerid:5,policyid:121,agentid:102,startDate:new Date("2024-11-01"),endDate:new Date("2025-12-01")},
  {assignmentid:207,customerid:5,policyid:120,agentid:102,startDate:new Date("2025-11-01"),endDate:new Date("2026-11-01")}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69575684fae8ace425523766'),
    '1': ObjectId('69575684fae8ace425523767'),
    '2': ObjectId('69575684fae8ace425523768'),
    '3': ObjectId('69575684fae8ace425523769'),
    '4': ObjectId('69575684fae8ace42552376a')
  }
}
```

Inserting into claims Collection

```
insuranceDB> db.claims.insertMany([
  {claimid:1000,assignmentid:202,claimDate:new
  Date("2020-10-
  01"),claimAmount:10000,claimStatus:"Rejected"},
  {claimid:1001,assignmentid:205,claimDate
  :new Date("2023-11-
  01"),claimAmount:20000,claimStatus:"Accepted"},
  {claimid:1002,assignmentid:206,claimDate:new Date("2025-12-
  01"),claimAmount:15000,claimStatus:"Accepted"},
  {claimid:1003,assignmentid:207,claimDate:new Date("2025-12-15"),claimAmount:40000,claimStatus:"Accepted"}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('695756b5fae8ace42552376b'),
    '1': ObjectId('695756b5fae8ace42552376c'),
    '2': ObjectId('695756b5fae8ace42552376d'),
    '3': ObjectId('695756b5fae8ace42552376e')
  }
}
```

SELECT

Display Claims Collection Data

```
insuranceDB> db.customers.find()
```

```
[
  {
    _id: ObjectId('695755fcfae8ace42552375a'),
    customerid: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    dateOfBirth: ISODate('1995-10-01T00:00:00.000Z'),
    phone: '99897275',
    email: 'amit@mail.com',
    address: null,
    city: null
  },
  {
    _id: ObjectId('695755fcfae8ace42552375b'),
    customerid: 2,
    firstName: 'preethi',
    lastName: 'varma',
    dateOfBirth: ISODate('1998-09-01T00:00:00.000Z'),
    phone: '89898989',
    email: 'preethi@mail.com',
    address: null,
    city: null
  },
  {
    _id: ObjectId('695755fcfae8ace42552375c'),
    customerid: 3,
    firstName: 'Vishwa',
    lastName: 'Nair',
    dateOfBirth: ISODate('2014-10-01T00:00:00.000Z'),
    phone: '89765987',
    email: 'vishwa@mail.com',
    address: null,
    city: null
  },
  {
    _id: ObjectId('695755fcfae8ace42552375d'),
    customerid: 4,
    firstName: 'Hansy',
    lastName: 'Kamtala',
    dateOfBirth: ISODate('2008-10-01T00:00:00.000Z'),
    phone: '83281187',
    email: 'hansika@mail.com',
    address: null,
    city: null
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    dateOfBirth: ISODate('2000-09-01T00:00:00.000Z'),
    phone: '99897678',
    email: 'julien@mail.com',
    address: null,
    city: null
  }
]
```

Display Policy Collection

insuranceDB> db.policies.find()

```
[
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyid: 120,
    policyName: 'Health Secure',
    policyType: 'Health',
    premiumAmount: 17569,
    durationYears: 1
  },
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyid: 121,
    policyName: 'Life Shield',
    policyType: 'Life',
    premiumAmount: 25000,
    durationYears: 10
  },
  {
    _id: ObjectId('69575663fae8ace425523765'),
    policyid: 122,
    policyName: 'Motor Protect',
    policyType: 'Motor',
    premiumAmount: 8000,
    durationYears: 1
  }
]
```

Display Policy Assignments Collection

insuranceDB> db.policyAssignments.find()

```
[
  {
    _id: ObjectId('69575684fae8ace425523766'),
    assignmentid: 201,
    customerid: 2,
    policyid: 120,
    agentid: 101,
    startDate: ISODate('2025-10-01T00:00:00.000Z'),
    endDate: ISODate('2025-10-01T00:00:00.000Z')
  },
  {
    _id: ObjectId('69575684fae8ace425523767'),
    assignmentid: 202,
    customerid: 1,
    policyid: 120,
    agentid: 101,
    startDate: ISODate('2020-10-01T00:00:00.000Z'),
    endDate: ISODate('2026-12-01T00:00:00.000Z')
  },
  {
    _id: ObjectId('69575684fae8ace425523768'),
    assignmentid: 205,
    customerid: 5,
    policyid: 122,
    agentid: 103,
    startDate: ISODate('2025-12-01T00:00:00.000Z'),
    endDate: ISODate('2026-12-01T00:00:00.000Z')
  },
  {
    _id: ObjectId('69575684fae8ace425523769'),
    assignmentid: 206,
    customerid: 5,
    policyid: 121,
    agentid: 102,
    startDate: ISODate('2024-11-01T00:00:00.000Z'),
    endDate: ISODate('2025-12-01T00:00:00.000Z')
  },
  {
    _id: ObjectId('69575684fae8ace42552376a'),
    assignmentid: 207,
    customerid: 5,
    policyid: 120,
    agentid: 102,
    startDate: ISODate('2025-11-01T00:00:00.000Z'),
    endDate: ISODate('2026-11-01T00:00:00.000Z')
  }
]
```

Display Agent Collection

insuranceDB> db.agents.find()

```
insuranceDB> db.agents.find()
[
  {
    _id: ObjectId('69575647fae8ace42552375f'),
    agentid: 101,
    agentName: 'Santhosh',
    phone: '99898989',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69575647fae8ace425523760'),
    agentid: 102,
    agentName: 'Ravi',
    phone: '97658987',
    city: 'Delhi'
  },
  {
    _id: ObjectId('69575647fae8ace425523761'),
    agentid: 103,
    agentName: 'Girish',
    phone: '89787897',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69575647fae8ace425523762'),
    agentid: 104,
    agentName: 'Hithesh',
    phone: '83383345',
    city: 'Kakinada'
  }
]
```

Using find with variations

findOne

insuranceDB> db.agents.findOne()

```
{
  _id: ObjectId('69575647fae8ace42552375f'),
  agentid: 101,
  agentName: 'Santhosh',
  phone: '99898989',
  city: 'Hyderabad'
}
```

find with ignoring fields

insuranceDB> db.agents.find({}, {agentid:0})

```
[
  {
    _id: ObjectId('69575647fae8ace42552375f'),
    agentName: 'Santhosh',
    phone: '99898989',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69575647fae8ace425523760'),
    agentName: 'Ravi',
    phone: '97658987',
    city: 'Delhi'
  },
  {
    _id: ObjectId('69575647fae8ace425523761'),
    agentName: 'Girish',
    phone: '89787897',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69575647fae8ace425523762'),
    agentName: 'Hithesh',
    phone: '83383345',
    city: 'Kakinada'
  }
]
```

Specific Fields to display

insuranceDB> db.agents.find({}, {agentid:1})

```
insuranceDB> db.agents.find({}, {agentid:1})
[
  { _id: ObjectId('69575647fae8ace42552375f'), agentid: 101 },
  { _id: ObjectId('69575647fae8ace425523760'), agentid: 102 },
  { _id: ObjectId('69575647fae8ace425523761'), agentid: 103 },
  { _id: ObjectId('69575647fae8ace425523762'), agentid: 104 }
]
```

Including/Excluding specific fields

insuranceDB> db.agents.find({}, {agentid:1, phone:0})

MongoServerError: Cannot do exclusion on field phone in inclusion projection

insuranceDB> db.agents.find({}, {agentid:1, _id:0})

```
insuranceDB> db.agents.find({}, {agentid:1, _id:0})
[
  { agentid: 101 },
  { agentid: 102 },
  { agentid: 103 },
  { agentid: 104 }
]
```

Update

insuranceDB> db.customers.updateOne({firstName:'Amit'}, { \$set:{address:"Hyderabad"}})

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Check whether updated or not with find command

insuranceDB> db.customers.findOne({firstName:'Amit'})

```
{
  _id: ObjectId('695755fcfae8ace42552375a'),
  customerid: 1,
  firstName: 'Amit',
  lastName: 'Sharma',
}
```

```
dateOfBirth: ISODate('1995-10-01T00:00:00.000Z'),
phone: '99897275',
email: 'amit@mail.com',
address: 'Hyderabad',
city: null
}
```

updateMany

```
insuranceDB> db.customers.updateMany({}, { $set: {address: "Hyderabad", city: "Hyderabad"}})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

```
insuranceDB> db.customers.find({})
```

```
[
  {
    _id: ObjectId('695755fcfae8ace42552375a'),
    customerid: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    dateOfBirth: ISODate('1995-10-01T00:00:00.000Z'),
    phone: '99897275',
    email: 'amit@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('695755fcfae8ace42552375b'),
    customerid: 2,
    firstName: 'preethi',
    lastName: 'varma',
    dateOfBirth: ISODate('1998-09-01T00:00:00.000Z'),
    phone: '89898989',
    email: 'preethi@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('695755fcfae8ace42552375c'),
    customerid: 3,
    firstName: 'Vishwa',
    lastName: 'Nain',
    dateOfBirth: ISODate('2014-10-01T00:00:00.000Z'),
    phone: '89765987',
    email: 'vishwa@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('695755fcfae8ace42552375d'),
    customerid: 4,
    firstName: 'Hansy',
    lastName: 'Kamtala',
    dateOfBirth: ISODate('2008-10-01T00:00:00.000Z'),
    phone: '83281187',
    email: 'hansika@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    dateOfBirth: ISODate('2000-09-01T00:00:00.000Z'),
    phone: '99897678',
    email: 'julien@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad'
  }
]
```


Delete

insuranceDB> db.agents.deleteOne({agentid:104})

```
{ acknowledged: true, deletedCount: 1 }
```

insuranceDB> db.agents.find({})

```
[
  {
    _id: ObjectId('69575647fae8ace42552375f'),
    agentid: 101,
    agentName: 'Santhosh',
    phone: '99898989',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69575647fae8ace425523760'),
    agentid: 102,
    agentName: 'Ravi',
    phone: '97658987',
    city: 'Delhi'
  },
  {
    _id: ObjectId('69575647fae8ace425523761'),
    agentid: 103,
    agentName: 'Girish',
    phone: '89787897',
    city: 'Hyderabad'
  }
]
```

Practice Queries

1. Display policies with PremiumAmount greater than 25,000.

insuranceDB> db.policies.find({premiumAmount:{\$gte:20000}})

```
[
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyid: 121,
    policyName: 'Life Shield',
    policyType: 'Life',
    premiumAmount: 25000,
    durationYears: 10
  }
]
```

2. Display policyName with PremiumAmount greater than 25,000.

insuranceDB> db.policies.find({premiumAmount:{\$gte:20000}},{policyName:1,_id:0})

```
[ { policyName: 'Life Shield' } ]
```

3. Write a query to find all customers who live in 'Delhi'.

insuranceDB> db.customers.find({address:'Hyderabad'},{firstName:1})

```
[
  { _id: ObjectId('695755fcfae8ace42552375a'), firstName: 'Amit' },
  { _id: ObjectId('695755fcfae8ace42552375b'), firstName: 'preethi' },
  { _id: ObjectId('695755fcfae8ace42552375c'), firstName: 'Vishwa' },
  { _id: ObjectId('695755fcfae8ace42552375d'), firstName: 'Hansy' },
  { _id: ObjectId('695755fcfae8ace42552375e'), firstName: 'Julien' }
]
```

4. Display policies that expire on or before '2025-12-31'.

```
db.policyAssignments.find({endDate:{$lt:new Date('2025-12-31')}})
```

```
[
  {
    _id: ObjectId('69575684fae8ace425523766'),
    assignmentid: 201,
    customerid: 2,
    policyid: 120,
    agentid: 101,
    startDate: ISODate('2025-10-01T00:00:00.000Z'),
    endDate: ISODate('2025-10-01T00:00:00.000Z')
  },
  {
    _id: ObjectId('69575684fae8ace425523769'),
    assignmentid: 206,
    customerid: 5,
    policyid: 121,
    agentid: 102,
    startDate: ISODate('2024-11-01T00:00:00.000Z'),
    endDate: ISODate('2025-12-01T00:00:00.000Z')
  }
]
```

5.Return policyTypes not equal to Motor

```
db.policies.find({policyType:{$ne:"Motor"}},{policyName:1})
```

```
[
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyName: 'Health Secure'
  },
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyName: 'Life Shield'
  }
]
```

6. Find policies with PolicyType IN (Motor, 'Health').

```
db.policies.find({policyType:{$in:["Motor","Health"]},{policyName:1})
```

```
insuranceDB> db.policies.find({policyType:{$in:["Motor","Health"]}}, {policyName:1})
[
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyName: 'Health Secure'
  },
  {
    _id: ObjectId('69575663fae8ace425523765'),
    policyName: 'Motor Protect'
  }
]
```

7. Find customers whose FirstName starts with 'A'.

```
insuranceDB> db.customers.find({ firstName: { $regex: "^A", $options: "i" } })
```

```
[
  {
    _id: ObjectId('695755fcfae8ace42552375a'),
    customerid: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    dateOfBirth: ISODate('1995-10-01T00:00:00.000Z'),
    phone: '99897275',
    email: 'amit@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad'
  }
]
```

8. Find policies which includes term Motor

```
insuranceDB> db.policies.find({ policyType: { $regex: "Motor", $options: "i" } })
```

```
[
  {
    _id: ObjectId('69575663fae8ace425523765'),
    policyid: 122,
    policyName: 'Motor Protect',
    policyType: 'Motor',
    premiumAmount: 8000,
    durationYears: 1
  }
]
```

9. Find total number of customers

```
insuranceDB> db.customers.aggregate({$count:"total_customers"})
```

```
[ { total_customers: 5 } ]
```

10. To get the SUM of premiumAmount

```
insuranceDB>
```

```
db.policies.aggregate([{$group:{_id:null,totalPremium:{$sum:"$premiumAmount"}}}])
```

```
[ { _id: null, totalPremium: 50569 } ]
```

11. Policies with premium between 10000 and 20000

```
insuranceDB> db.policies.find({premiumAmount:{$gt:10000,$lt:30000}})
```

```
[
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyid: 120,
    policyName: 'Health Secure',
    policyType: 'Health',
    premiumAmount: 17569,
    durationYears: 1
  },
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyid: 121,
    policyName: 'Life Shield',
    policyType: 'Life',
    premiumAmount: 25000,
    durationYears: 10
  }
]
```

12. Policies sorted by premium

```
insuranceDB> db.policies.find().sort({premiumAmount:-1})
```

```
[
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyid: 121,
    policyName: 'Life Shield',
    policyType: 'Life',
    premiumAmount: 25000,
    durationYears: 10
  },
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyid: 120,
    policyName: 'Health Secure',
    policyType: 'Health',
    premiumAmount: 17569,
    durationYears: 1
  },
  {
    _id: ObjectId('69575663fae8ace425523765'),
    policyid: 122,
    policyName: 'Motor Protect',
    policyType: 'Motor',
    premiumAmount: 8000,
    durationYears: 1
  }
]
```

13. Top 2 highest premium policies

insuranceDB> db.policies.find().sort({premiumAmount:-1}).limit(2)

```
[
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyid: 121,
    policyName: 'Life Shield',
    policyType: 'Life',
    premiumAmount: 25000,
    durationYears: 10
  },
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyid: 120,
    policyName: 'Health Secure',
    policyType: 'Health',
    premiumAmount: 17569,
    durationYears: 1
  }
]
```

14. Show policyType in uppercase

db.policies.aggregate([{ \$project: { policyTypeUpper: { \$toUpper: "\$policyType" } } }])

```
[
  {
    _id: ObjectId('69575663fae8ace425523763'),
    policyTypeUpper: 'HEALTH'
  },
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyTypeUpper: 'LIFE'
  },
  {
    _id: ObjectId('69575663fae8ace425523765'),
    policyTypeUpper: 'MOTOR'
  }
]
```

15. Add 18% GST to premium

insuranceDB>

db.policies.aggregate([{\$project:{firstName:1,totalPremium:{"\$multiply":["\$premiumAmount",1.18]}}}])

```
[
  { _id: ObjectId('69575663fae8ace425523763'), totalPremium: 20731.42 },
  { _id: ObjectId('69575663fae8ace425523764'), totalPremium: 29500 },
  { _id: ObjectId('69575663fae8ace425523765'), totalPremium: 9440 }
]
```

16. To get the number of policies per durationYears

```
insuranceDB> db.policies.aggregate([{$group:{_id:"$durationYears",totalPolicies:{$sum:1}}}]
```

```
[ { _id: 1, totalPolicies: 2 }, { _id: 10, totalPolicies: 1 } ]
```

17. Highest premium per policyType

```
insuranceDB>
```

```
db.policies.aggregate([{$group:{_id:"$policyType",maxPremium:{$max:"$premiumAmount"}}}]
```

```
[
  { _id: 'Life', maxPremium: 25000 },
  { _id: 'Health', maxPremium: 17569 },
  { _id: 'Motor', maxPremium: 8000 }
]
```

18. Only policies with more than 1 policyAssignment

```
insuranceDB>
```

```
db.policyAssignments.aggregate([{$group:{_id:"$policyid",count:{$sum:1}}},{ $match:{count:{$gt:1}}}]
```

```
[ { _id: 120, count: 3 } ]
```

19. Sort policy types by average premium

```
insuranceDB> db.policies.aggregate([ { $group: { _id: "$policyType", avgPremium: { $avg: "$premiumAmount" } } }, { $sort: { avgPremium: -1 } } ] )
```

```
[
  { _id: 'Life', avgPremium: 25000 },
  { _id: 'Health', avgPremium: 17569 },
  { _id: 'Motor', avgPremium: 8000 }
]
```

20. Increase all Health premiums by 10%

```
insuranceDB> db.policies.updateMany( { policyType: "Health" }, { $mul: { premium: 1.1 } } )
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
db.policies.deleteMany({ premium: { $gt: 25000 } })
```

Joins

```
insuranceDB> db.customers.aggregate([ { $lookup: { from: "policies", localField:
"customerId", foreignField: "customerId", as: "policyDetails" } } ])
```

```
{
  _id: ObjectId('695755fcfae8ace42552375d'),
  customerId: 4,
  firstName: 'Hansy',
  lastName: 'Kamtala',
  dateOfBirth: ISODate('2008-10-01T00:00:00.000Z'),
  phone: '83281187',
  email: 'hansika@mail.com',
  address: 'Hyderabad',
  city: 'Hyderabad',
  policyDetails: [
    {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    },
    {
      _id: ObjectId('69575663fae8ace425523764'),
      policyid: 121,
      policyName: 'Life Shield',
      policyType: 'Life',
      premiumAmount: 25000,
      durationYears: 10
    },
    {
      _id: ObjectId('69575663fae8ace425523765'),
      policyid: 122,
      policyName: 'Motor Protect',
      policyType: 'Motor',
      premiumAmount: 8000,
      durationYears: 1
    }
  ]
}
```

23. All customers, even without policies

insuranceDB> db.customers.aggregate([{ \$lookup: { from: "policyAssignments", localField: "customerid", foreignField: "customerid", as: "assignments" } }, { \$lookup: { from: "policies", localField: "assignments.policyid", foreignField: "policyid", as: "policyDetails" } }, { \$project: { customerid: 1, firstName: 1, policyDetails: 1, _id: 0 } }])

```
[
  {
    customerid: 1,
    firstName: 'Amit',
    policyDetails: [
      {
        _id: ObjectId('69575663fae8ace425523763'),
        policyid: 120,
        policyName: 'Health Secure',
        policyType: 'Health',
        premiumAmount: 17569,
        durationYears: 1,
        premium: 0
      }
    ]
  },
  {
    customerid: 2,
    firstName: 'preethi',
    policyDetails: [
      {
        _id: ObjectId('69575663fae8ace425523763'),
        policyid: 120,
        policyName: 'Health Secure',
        policyType: 'Health',
        premiumAmount: 17569,
        durationYears: 1,
        premium: 0
      }
    ]
  },
  {
    customerid: 3, firstName: 'Vishwa', policyDetails: [] },
  {
    customerid: 4, firstName: 'Hansy', policyDetails: [] },
  {
    customerid: 5,
    firstName: 'Julien',
    policyDetails: [
      {
        _id: ObjectId('69575663fae8ace425523763'),
        policyid: 120,
        policyName: 'Health Secure',
        policyType: 'Health',
        premiumAmount: 17569,
        durationYears: 1,
        premium: 0
      }
    ]
  }
]
```

```
    premium: 0
  },
  {
    _id: ObjectId('69575663fae8ace425523764'),
    policyid: 121,
    policyName: 'Life Shield',
    policyType: 'Life',
    premiumAmount: 25000,
    durationYears: 10
  },
  {
    _id: ObjectId('69575663fae8ace425523765'),
    policyid: 122,
    policyName: 'Motor Protect',
    policyType: 'Motor',
    premiumAmount: 8000,
    durationYears: 1
  }
]
insuranceDB>
```


24. Customers with policies (Join 3 tables)

insuranceDB> db.customers.aggregate([{ \$lookup: { from: "policyAssignments", localField: "customerid", foreignField: "customerid", as: "assignments" } }, { \$lookup: { from: "policies", localField: "assignments.policyid", foreignField: "policyid", as: "policyDetails" } }, { \$project: { customerid: 1, firstName: 1, policyDetails: 1, _id: 0 } }, { \$unwind: "\$policyDetails" }])

```
[
  {
    customerid: 1,
    firstName: 'Amit',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    }
  },
  {
    customerid: 2,
    firstName: 'preethi',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    }
  },
  {
    customerid: 5,
    firstName: 'Julien',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    }
  },
  {
    customerid: 5,
    firstName: 'Julien',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523764'),
      policyid: 121,
      policyName: 'Life Shield',
      policyType: 'Life',
      premiumAmount: 25000,
      durationYears: 10
    }
  },
  {
    customerid: 5,
    firstName: 'Julien',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523765'),
      policyid: 122,
      policyName: 'Motor Protect',
      policyType: 'Motor',
      premiumAmount: 8000,
      durationYears: 1
    }
  }
]
```

25. Customers with Health policy

```
insuranceDB> db.customers.aggregate([ { $lookup: { from: "policyAssignments", localField:
"customerid", foreignField: "customerid", as: "assignments" } }, { $lookup: { from: "policies",
localField: "assignments.policyid", foreignField: "policyid", as: "policyDetails" } }, { $project: {
customerid: 1, firstName: 1, policyDetails: 1, _id: 0 } }, { $unwind: "$policyDetails"
},{ $match: {"policyDetails.policyType": "Health" } } ] )
```

```
[
  {
    customerid: 1,
    firstName: 'Amit',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    }
  },
  {
    customerid: 2,
    firstName: 'preethi',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    }
  },
  {
    customerid: 5,
    firstName: 'Julien',
    policyDetails: {
      _id: ObjectId('69575663fae8ace425523763'),
      policyid: 120,
      policyName: 'Health Secure',
      policyType: 'Health',
      premiumAmount: 17569,
      durationYears: 1,
      premium: 0
    }
  }
]
```

26. Customers with policies (Join with 2 tables)

insuranceDB> db.customers.aggregate([{ \$lookup: { from: "policyAssignments", localField: "customerid", foreignField: "customerid", as: "policyDetails" } }, { \$unwind: "\$policyDetails" }, { \$project: { customerid: 1, firstName: 1, lastName: 1, policyDetails: 1 } }])

```
[
  {
    _id: ObjectId('695755fcfae8ace42552375a'),
    customerid: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    policyDetails: {
      _id: ObjectId('69575684fae8ace425523767'),
      assignmentid: 202,
      customerid: 1,
      policyid: 120,
      agentid: 101,
      startDate: ISODate('2020-10-01T00:00:00.000Z'),
      endDate: ISODate('2026-12-01T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375b'),
    customerid: 2,
    firstName: 'preethi',
    lastName: 'varma',
    policyDetails: {
      _id: ObjectId('69575684fae8ace425523766'),
      assignmentid: 201,
      customerid: 2,
      policyid: 120,
      agentid: 101,
      startDate: ISODate('2025-10-01T00:00:00.000Z'),
      endDate: ISODate('2025-10-01T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    policyDetails: {
      _id: ObjectId('69575684fae8ace425523768'),
      assignmentid: 205,
      customerid: 5,
      policyid: 122,
      agentid: 103,
      startDate: ISODate('2025-12-01T00:00:00.000Z'),
      endDate: ISODate('2026-12-01T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    policyDetails: {
      _id: ObjectId('69575684fae8ace425523769'),
      assignmentid: 206,
      customerid: 5,
      policyid: 121,
      agentid: 102,
      startDate: ISODate('2024-11-01T00:00:00.000Z'),
      endDate: ISODate('2025-12-01T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    policyDetails: {
      _id: ObjectId('69575684fae8ace42552376a'),
      assignmentid: 207,
      customerid: 5,
      policyid: 120,
      agentid: 102,
      startDate: ISODate('2025-11-01T00:00:00.000Z'),
      endDate: ISODate('2026-11-01T00:00:00.000Z')
    }
  }
]
```

27. .Customers with policies(Join with 2 tables limiting columns)

```
[
  {
    _id: ObjectId('695755fcfae8ace42552375a'),
    customerid: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    policyDetails: { policyid: 120 }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375b'),
    customerid: 2,
    firstName: 'preethi',
    lastName: 'varma',
    policyDetails: { policyid: 120 }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    policyDetails: { policyid: 122 }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    policyDetails: { policyid: 121 }
  },
  {
    _id: ObjectId('695755fcfae8ace42552375e'),
    customerid: 5,
    firstName: 'Julien',
    lastName: 'Gou',
    policyDetails: { policyid: 120 }
  }
]
```

28. Customers with no policies

insuranceDB> db.customers.aggregate([{ \$lookup: { from: "policyAssignments", localField: "customerid", foreignField: "customerid", as: "policies" } }, { \$match: { policies: { \$size: 0 } } }])

```
[
  {
    _id: ObjectId('695755fcfae8ace42552375c'),
    customerid: 3,
    firstName: 'Vishwa',
    lastName: 'Nair',
    dateOfBirth: ISODate('2014-10-01T00:00:00.000Z'),
    phone: '89765987',
    email: 'vishwa@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad',
    policies: []
  },
  {
    _id: ObjectId('695755fcfae8ace42552375d'),
    customerid: 4,
    firstName: 'Hansy',
    lastName: 'Kamtala',
    dateOfBirth: ISODate('2008-10-01T00:00:00.000Z'),
    phone: '83281187',
    email: 'hansika@mail.com',
    address: 'Hyderabad',
    city: 'Hyderabad',
    policies: []
  }
]
```